

User Guide for MIDAS

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Abstract

Short introduction to subject of the paper ...

1 Introduction

2 3D full wave model for scattering by complex-shaped objects

In order to rigorously compute the scattering properties of complex snow scatterer, we develop a 3D full-wave model based on the volumetric integral representation of the electric field, and we apply it to complex-shaped scatterer.

2.1 Modeling of the electric field inside the scatterer based on the volumetric integral formulation

2.1.1 Volume Integral Equation Method (VIEM)

The model developed to calculate the total field inside complex-shaped hydrometeors is based on the use of the volume integral equation method (VIEM) [1, 2], where the electric field $\vec{E}(\vec{r})$ at any point $\vec{r}$ inside the scatterer is given by

$$\vec{E}(\vec{r}) = \vec{E}^i(\vec{r}) + (k_0^2 + \nabla \nabla \cdot) \int_{\Omega} \chi(\vec{r}') \bar{\bar{G}}(\vec{r}, \vec{r}') \vec{E}(\vec{r}') d\vec{r}' \quad (1)$$

where $\vec{E}(\vec{r})$ is the field inside the scatterer, $\vec{E}^i(\vec{r})$ is the incident field, k_0 is the wave-length number in the air, Ω represents the domain occupied by the scatterer, $\chi(\vec{r}')$ is the dielectric contrast at the location $\vec{r}'$ and $\bar{\bar{G}}(\vec{r}, \vec{r}')$ is the free space dyadic Green's function.

In electromagnetic theory, the dyadic Green's function $\bar{\bar{G}}(\vec{r}, \vec{r}')$ is essentially defined by the scattered field $\vec{E}$ at any observation point $\vec{r}$ generated by a radiating electric dipole $\vec{p}$ located at the source position $\vec{r}'$. In mathematical terms this reads as

$$\vec{\mathbf{E}}(\vec{\mathbf{r}}) = \omega^2 \mu_0 \mu \bar{\bar{\mathbf{G}}}(\vec{\mathbf{r}}, \vec{\mathbf{r}}') \vec{\mathbf{p}}(\vec{\mathbf{r}}') \quad (2)$$

It verifies the equation

$$(\nabla^2 + k_0^2) \bar{\bar{\mathbf{G}}}(\vec{\mathbf{r}}, \vec{\mathbf{r}}') = -\delta(\vec{\mathbf{r}} - \vec{\mathbf{r}}') \underline{\mathbf{I}} \quad (3)$$

where δ is the Dirac delta function and $\underline{\mathbf{I}}$ is the identity. The dyadic green's function in free space is expressed as

$$\bar{\bar{\mathbf{G}}}(\vec{\mathbf{r}}, \vec{\mathbf{r}}') = \begin{bmatrix} G_{xx} & 0 & 0 \\ 0 & G_{yy} & 0 \\ 0 & 0 & G_{zz} \end{bmatrix} \quad (4)$$

$$\mathbf{G}_{pq}(\vec{\mathbf{r}}, \vec{\mathbf{r}}') = \frac{e^{jk_0|\vec{\mathbf{r}}-\vec{\mathbf{r}}'|}}{4\pi|\vec{\mathbf{r}}-\vec{\mathbf{r}}'|} \quad (5)$$

where $p, q = x, y, z$. It is worth noting that the dyadic green's function in free space is expressed in the spectral domain as

$$\bar{\bar{\mathbf{G}}}(\vec{\mathbf{r}}, \vec{\mathbf{r}}') = \int \int_{-\infty}^{+\infty} g_{pq}^s(\nu, \eta, z, z') e^{j2\pi[\nu(x-x')+\eta(y-y')]} d\nu d\eta \quad (6)$$

$$g_{pq}^s(\nu, \eta, z, z') = \frac{j}{2\gamma_0} e^{j\gamma_0|z-z'|} \quad (7)$$

where $p, q = x, y, z$, $\gamma_0 = \sqrt{k_0^2 - 4\pi^2(\nu^2 + \eta^2)}$, $\nu = \frac{k_{0x}}{2\pi}$, $\eta = \frac{k_{0y}}{2\pi}$, and k_{0x} and k_{0y} are the components of the free space wave vector k_0 along x and y , respectively.

2.1.2 Calculation of the Incident Field $\vec{\mathbf{E}}^i(\vec{\mathbf{r}})$

We consider an incident plane wave propagating along $\hat{k}_i = \frac{\vec{\mathbf{k}}_i}{k_0}$, polarized vertically or horizontally, depending if $\hat{e}_i = \hat{v}_i$ or $\hat{e}_i = \hat{h}_i$, respectively, as described in "here reference to Brngi". $\hat{k}_i$ is set by the angles θ_i and ϕ_i where, depending on the used convention (DDScat- or ADDA-equivalent respectively), θ_i describes the rotation around $\hat{z}$ or $\hat{y}$, and ϕ_i describes the rotation around $\hat{x}$ or $\hat{z}$. (see figures - Not included yet for ADDA convention). Knowing that any observation point is described in the Cartesian reference as $\vec{\mathbf{r}} = x\hat{x} + y\hat{y} + z\hat{z}$, the incident electric field, $\vec{\mathbf{E}}^i(\vec{\mathbf{r}})$, is expressed at any observation point $\vec{\mathbf{r}}$ as

$$\vec{\mathbf{E}}^i(\vec{\mathbf{r}}) = \hat{e}_i E_0 e^{jk_0 \hat{k}_i \vec{\mathbf{r}}} \quad (8)$$

Depending on the used convention, DDScat- (Eq. 9) or ADDA- (Eq. 10), the wave and polarization unit vectors ($\hat{k}_i$, $\hat{v}_i$, $\hat{h}_i$) are expressed as

$$\begin{bmatrix} \hat{k}_i \\ \hat{v}_i \\ \hat{h}_i \end{bmatrix} = \begin{bmatrix} \cos(\theta_i) & \sin(\theta_i) * \cos(\phi_i) & \sin(\theta_i) * \sin(\phi_i) \\ -\sin(\theta_i) & \cos(\theta_i) * \cos(\phi_i) & \cos(\theta_i) * \sin(\phi_i) \\ 0 & -\sin(\phi_i) & \cos(\phi_i) \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{bmatrix} \quad (9)$$

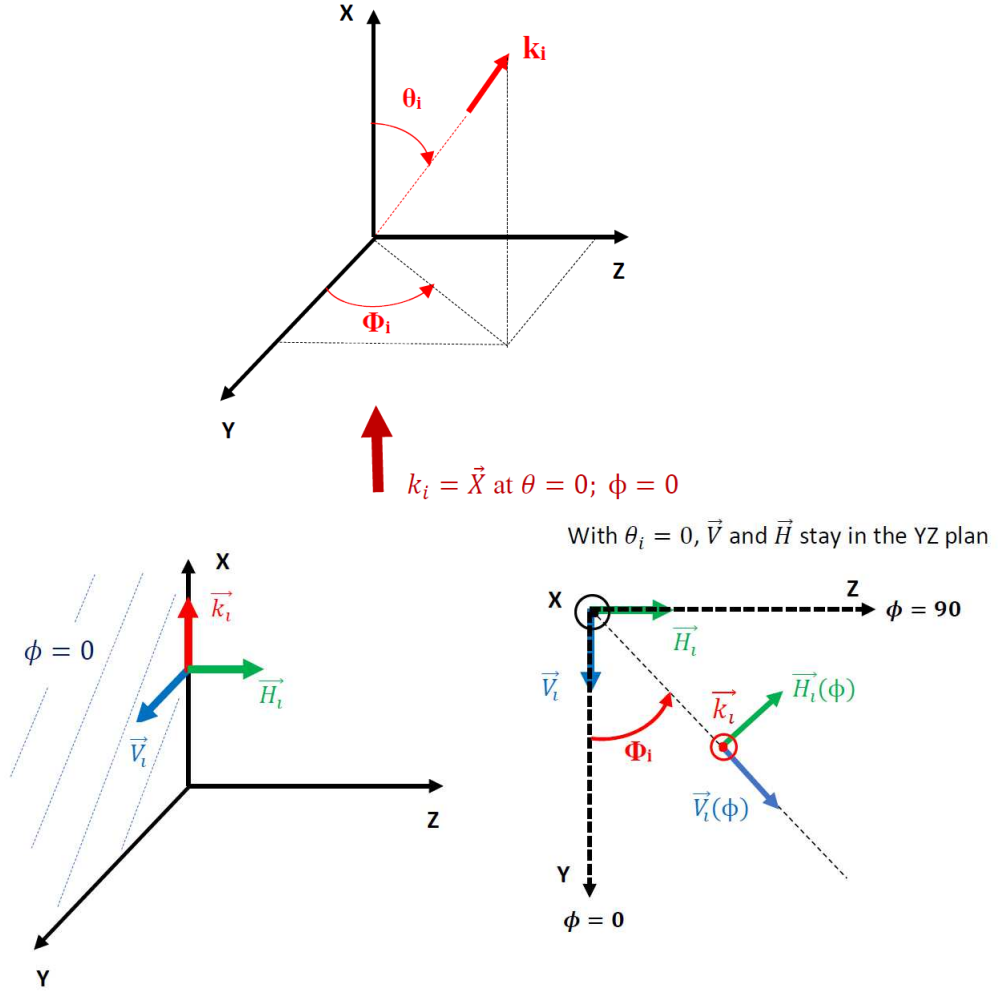


Figure 1: Incident EM wave direction in DDScat convention (Eq. 9).

$$\begin{bmatrix} \hat{k}_i \\ \hat{v}_i \\ \hat{h}_i \end{bmatrix} = \begin{bmatrix} \sin(\theta_i) * \cos(\phi_i) & \sin(\theta_i) * \sin(\phi_i) & \cos(\theta_i) \\ \cos(\theta_i) * \cos(\phi_i) & \cos(\theta_i) * \sin(\phi_i) & -\sin(\theta_i) \\ -\sin(\phi_i) & \cos(\phi_i) & 0 \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{bmatrix} \quad (10)$$

2.1.3 The Method of Moments

Next, the volumetric integral equation is solved by means of a Method of Moments (MoM). To do so, Eq. 1 is rewritten as

$$\bar{\Gamma} \vec{E}(\vec{r}) = \vec{E}^i(\vec{r}) \quad (11)$$

$$\text{where } \bar{\Gamma} = \bar{\mathbf{I}} - (k_0^2 + \nabla \nabla \cdot) \int_{\Omega} \chi(\vec{\mathbf{r}}') \bar{\mathbf{G}}(\vec{\mathbf{r}}, \vec{\mathbf{r}}') d\vec{\mathbf{r}}'$$

The MoM begins with the discretization of object into N cubic cells Ω_n , of side S_c , which is small enough to enable us to assume that the electric field inside is constant. To meet this requirement, and thus to ensure the accuracy of the MoM solution, the cell size should be equal to or lower than $\lambda_s/10$, with $\lambda_s = \lambda_0/\sqrt{\epsilon_r}$. λ_s is the wavelength inside the scatterer, λ_0 is the free space wavelength and ϵ_r is the real part of the relative permittivity of the discretized domain. Fig. 2 shows an example of discretization of a simple spherical particle and of 3 typical snow crystals : a star-shaped dendrite, a dendrite with broadening arms and a sandwich plate.

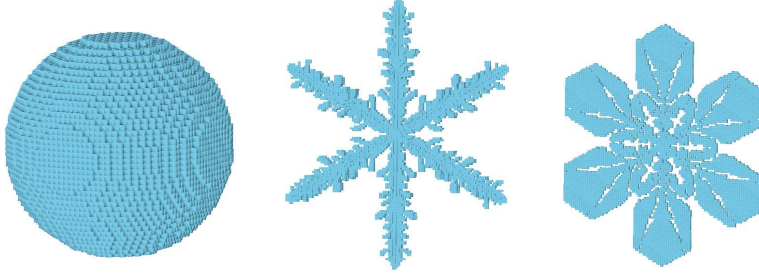


Figure 2: Examples of discretization into cubic cells (or voxels).

After discretization into N cells, the field inside the scatterer is decomposed as

$$\vec{\mathbf{E}}(\vec{\mathbf{r}}) = \sum_{n=1}^N \sum_{q=1}^3 E_q^n \bar{\mathbf{F}}_q^n(\vec{\mathbf{r}}) \quad (12)$$

where E_q^n is the constant unknown of $\bar{\mathbf{F}}_q^n$, the n^{th} basis function for the component ($q = x, y$ or z) of the field inside the scatterer. To select a set of test functions $\bar{\mathbf{W}}_p^m$ ($m = 1, \dots, M$ and $p = x, y, z$), the point matching method is used. So, $M = N$ and $\bar{\mathbf{W}}_p^m$ is a Dirac delta function located at the center of the cell Ω_n . Eq. 11 is re-written :

$$\sum_{n=1}^N \sum_{q=1}^3 \langle \bar{\mathbf{W}}_p^m, \bar{\Gamma} \bar{\mathbf{F}}_q^n \rangle E_q^n = \langle \bar{\mathbf{W}}_p^m, \bar{\mathbf{E}}^i \rangle \quad (13)$$

$$\text{Where } \bar{\mathbf{F}}_q^n(\vec{\mathbf{r}}) = \begin{cases} 1; & \vec{\mathbf{r}} \in \Omega_n \\ 0; & \vec{\mathbf{r}} \notin \Omega_n \end{cases} \hat{q}$$

$$\text{and } \bar{\mathbf{W}}_p^m(\vec{\mathbf{r}}) = \delta(\vec{\mathbf{r}} - \vec{\mathbf{r}}_m) \hat{p}$$

This results in the system of N linear equations

$$\sum_{n=1}^N \sum_{q=1}^3 \mathbf{Z}_{pq}^{mn} \mathbf{E}_q^n = \mathbf{E}_p^{i,m} \quad (14)$$

that we can simply express as

$$\mathbf{Z}\mathbf{E} = \mathbf{E}^i \quad (15)$$

where $\mathbf{Z}$ is the $3N \times 3N$ full matrix representing the interactions between the cells composing the scatterer, $\mathbf{E}^i$ is the incident field of size $3N$ and $\mathbf{E}$ is the unknown solution vector of size $3N$ that represents the total electric field inside the scatterer in the q ($=x, y$ or z) direction.

The elements of $\mathbf{Z} = [Z_{pq}^{mn}]$ are given by $Z_{pq}^{mn} = \delta_{mn}\delta_{pq} - Z_{pq}^{f,mn}$ where

$$Z_{pq}^{f,mn} = k_0^2 \int_{\Omega_n} \chi(\mathbf{r}'_n) G_{pq}^f(\mathbf{r}_m, \mathbf{r}'_n) d\mathbf{r}'_n + \sum_{q=1}^3 \frac{\partial^2}{\partial p_m \partial q_m} \int_{\Omega_n} \chi(\mathbf{r}'_n) G_{pq}^f(\mathbf{r}_m, \mathbf{r}'_n) d\mathbf{r}'_n \quad (16)$$

where $\mathbf{r}_m$ points to the observation point m , and $\mathbf{r}'_n$ points to the center of the source cell n , G_{pq}^f is the component pq of the dyadic Greens function given by Eq. 5, and $\chi(\mathbf{r}') = \frac{\epsilon(\mathbf{r}') - \epsilon_0}{\epsilon_0}$ is the dielectric contrast. Given that the domain Ω_n is homogeneous, $\chi(\mathbf{r}')$ does not depend on the position $\mathbf{r}'$. We can then write :

$$\int_{\Omega_n} \chi(\mathbf{r}'_n) G_{pq}^f(\mathbf{r}_m, \mathbf{r}'_n) d\mathbf{r}'_n = \chi_n \int_{\Omega_n} G_{pq}^f(\mathbf{r}_m, \mathbf{r}'_n) d\mathbf{r}'_n \quad (17)$$

where $\chi_n = \epsilon_{r,n} - 1$

If $m = n$, in order to avoid singularities, $Z_{pq}^{f,mn}$ is computed using Hadamard regularization. The integral on Ω_n is approximated by an integral on sphere of radius

$$a_n = S_c \sqrt[3]{3/(4\pi)} \quad (18)$$

So, we obtain the following

$$\begin{aligned} \text{If } m = n : Z_{pq}^{f,mn} &= \mathcal{H} \left(\int_{\Omega_n} (k_0^2 + \frac{\partial^2}{\partial p_m^2}) G_{pq}^f(\mathbf{r}_m, \mathbf{r}'_n) d\mathbf{r}'_n \right) \chi_n \\ &= \left(\frac{2}{3} e^{jk_0 a_n} (1 - jk_0 a_n) - 1 \right) \chi_n & \text{if } p = q \\ Z_{pq}^{f,mn} &= 0 & \text{if } p \neq q \end{aligned} \quad (19)$$

With $m \neq n$, we can integrate numerically the elements of the dyadic Green's function over the domain Ω_n . But, this will result in a computational cost. So, we have chosen to approximate the integration of $G_{pq}^f(\mathbf{r}_m, \mathbf{r}'_n)$ over a cubical cell of side size S_c by an integration over a sphere of equivalent radius a_n (Eq. 18) as follows

$$\begin{aligned} \int_{\Omega_n} G_{pq}^f(\mathbf{r}_m, \mathbf{r}'_n) d\mathbf{r}'_n &= \kappa_n G_{pq}^f(\mathbf{r}_m, \mathbf{r}'_n) \\ \text{where } \kappa_n &= 4\pi \frac{\sin(k_0 a_n) - k_0 a_n \cos(k_0 a_n)}{k_0^3} \end{aligned} \quad (20)$$

Thus, the elements of $Z_{pq}^{f,mn}$, with $m \neq n$, are computed as follows

$$Z_{pq}^{f,mn} = \begin{cases} \frac{e^{jk_0 r_{mn}}}{4\pi r_{mn}^2} \left(\tau_{mn} + k_0^2 \left(r_{mn} - \frac{(p_m - q'_n)^2}{r_{mn}} \right) - 3 \frac{(p_m - q'_n)^2}{r_{mn}^2} \tau_{mn} \right), & m \neq q, p = q \\ \frac{e^{jk_0 r_{mn}} (p_m - p'_n)(q_m - q'_n)}{4\pi r_{mn}^3} \left(-k_0^2 - \frac{3}{r_{mn}} \tau_{mn} \right) \kappa_n, & m \neq q, p \neq q \end{cases} \quad (21)$$

where $\tau_{mn} = jk_0 - \frac{1}{r_{mn}}$, and $r_{mn} = \sqrt{(x_m - x'_n)^2 + (y_m - y'_n)^2 + (z_m - z'_n)^2}$ is the distance between the center of the observation cell m and the center of the source cell n .

Further details about the free space Green's function, the selection of basis and test functions and the calculation of the MoM matrix elements are available in [1, 2]. Since our 3D full-wave model is based on the use of piecewise constant basis functions with the VIEM, it is mathematically close to the DDA [3].

To summarize, an application of a conventional MoM to a complex-shaped scatterer discretized into N_c cubic cells results in a system of linear equations corresponding to a full matrix $\mathbf{Z}^{MoM}$ of size $3N_c \times 3N_c$. Therefore, with a complexity of $O(N^3)$ in time and of $O(N^2)$ in storage, where $N = 3N_c$, the model is so computationally extensive that it becomes very quickly untenable as we increase N . That is exactly the type of scenario where the CBFM is well suited to solve the problem at hand in an efficient manner.

2.2 Calculated scattering quantities

Once $\vec{\mathbf{E}}(\vec{\mathbf{r}})$ the total electric field inside the object is known, $\vec{\mathbf{E}}^s(\vec{\mathbf{r}})$ the scattered field at any point $\vec{\mathbf{r}}$ is given by

$$\vec{\mathbf{E}}^s(\vec{\mathbf{r}}) = (k_0^2 + \nabla \nabla \cdot) \int_{\Omega} \chi(\vec{\mathbf{r}}') \vec{\mathbf{G}}(\vec{\mathbf{r}}, \vec{\mathbf{r}}') \vec{\mathbf{E}}(\vec{\mathbf{r}}') d\vec{\mathbf{r}}' \quad (22)$$

Example :

Similarly to the calculation of the electric field inside the scatterer, the discretization of this integral equation leads to the following matrix system

$$\begin{aligned} [\mathbf{D}_{pq}^{s,n}] [\mathbf{E}_q^{n,i}] &= [\mathbf{E}_q^{s,n,i}] \\ \text{with } q &= [x, y, z]; n = 1, \dots, N_c \\ i &= 1, \dots, N_{inc} \text{ and } s = 1, \dots, N_{sca} \end{aligned} \quad (23)$$

where $\mathbf{D}$ is a $3N_{sca} \times 3N_c$ matrix representing the 3 components x, y and z of the scattered field from each cell $n \leq N_{bc}$ in the observation point $1 \leq s \leq N_{sca}$, $\mathbf{E}$ of size $3N_{bc} \times N_{inc}$ is the total electric field inside the scatterer calculated for each incident EM wave $1 \leq i \leq N_{inc}$, and $\mathbf{E}^s$ of size $N_{sca} \times N_{inc}$ represents the scattered field obtained at the observation points s with the incident EM waves i . Therefore, once the

electric field inside the scatterer is known for the N_{inc} incident directions, the calculation of the scattered field at the N_{sca} observation points scales as $O(N_c N_{inc} N_{sca})$. Note also that the calculation of $\mathbf{E}^s$ is embarrassingly parallel and so can be easily accelerated by the use of a large number of MPI processes.

Next, the scattering matrix $\mathbf{S}$ for each incident direction is derived from the scattered fields [4], and the extinction, scattering and back-scattering efficiency factors $Q_{ext} = C_{ext}/\pi a^2$, $Q_{scat} = C_{scat}/\pi a^2$ and $Q_{bks} = C_{bks}/\pi a^2$ are calculated. C_{ext} , C_{scat} and C_{bks} are the scattering cross-sections and a represents the effective radius of a particle of volume V , which is defined by $a = (3V/4\pi)^{1/3} = (3N_c/4\pi)^{1/3} S_c$ where S_c is the size of cell used to discretize the particle. For each incident direction defined by (θ_i, Φ_i) , C_{ext} , C_{scat} and C_{bks} are derived from the scattering matrix S as follows [4]

$$C_{ext} = \frac{2\pi}{k} [\Im\{\mathbf{S}_{VV}(\theta_{fwd}, \Phi_{fwd})\} + \Im\{\mathbf{S}_{HH}(\theta_{fwd}, \Phi_{fwd})\}] \quad (24)$$

$$C_{scat} = \frac{1}{2k} \int_0^{2\pi} \int_0^\pi [\mathbf{S}_{VV}(\theta_s, \Phi_s)^2 + \mathbf{S}_{VH}(\theta_s, \Phi_s)^2 + \mathbf{S}_{HV}(\theta_s, \Phi_s)^2 + \mathbf{S}_{HH}(\theta_s, \Phi_s)^2] \sin \theta d\theta d\Phi \quad (25)$$

$$C_{bks} = \frac{1}{2k} [\mathbf{S}_{VV}(\theta_{bks}, \Phi_{bks})^2 + \mathbf{S}_{VH}(\theta_{bks}, \Phi_{bks})^2 + \mathbf{S}_{HV}(\theta_{bks}, \Phi_{bks})^2 + \mathbf{S}_{HH}(\theta_{bks}, \Phi_{bks})^2] \quad (26)$$

where (θ_s, Φ_s) describes the scattering direction, and $(\theta_{fwd}, \Phi_{fwd})$ and $(\theta_{bks}, \Phi_{bks})$ refers respectively to the scattering-forward and the backscattering directions.

Next, an averaging of the scattered properties is performed over the calculated incident directions defined by (θ_i, ϕ_i) , as follows

$$\langle Q \rangle = \frac{1}{\Omega} \int_0^{\phi_m} \int_0^{\theta_m} Q(\theta, \phi) \sin \theta d\theta d\phi_i \quad (27)$$

where the choice of $0 \leq \phi_m \leq 2\pi$ and $0 \leq \theta_m \leq \pi$ strongly depends on the geometry of the snow particle, particularly on the symmetries that it presents. The solid angle Ω is calculated as $\int_0^{\phi_m} d\phi \int_0^{\theta_m} \sin \theta d\theta$. For the DDScat code, orientationally averaged scattering quantities are computed by evaluating them over a certain number of particle orientations. Thus, DDScat computes the orientational average $\langle Q \rangle$ of a quantity $Q(\beta, \theta, \phi)$ [5] as

$$\langle Q \rangle = \frac{1}{8\pi^2} \int_0^{2\pi} d\beta \int_{-1}^1 d\cos \theta \int_0^{2\pi} d\phi Q(\beta, \theta, \phi) \quad (28)$$

where the Euler angles $(\beta, \theta$ and $\phi)$ are used to describe the 3D orientation of the snow particle. The DDScat code also enables the user to take advantage of the symmetries presented by the simulated particle by choosing the range of values of β , θ and ϕ [5]. Since it is based on iterative solver, the computational cost for DDScat increases very rapidly with the number of target orientations, particularly in the case of electrically

large particles. This is due to the recalculation of the entire EM problem for each target orientation. On the other hand, using a direct solver-based method, the NESCoP code is expected to largely outperform DDScat in terms of CPU time when computing orientationally averaged scattering by electrically large snow particles with a high number of incident directions (abbreviated as **id** for the MoM/CBFM code) or target orientations (abbreviated as **to** for DDScat).

3 Conclusions

There is no longer L^AT_EX example which was written by [?].

References

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